

VU HOANG NGUYEN

Embedded Engineer | Fully Funded Erasmus Mundus Scholar

Currently in Norway on a post-study work permit; available to start immediately.

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ABOUT ME

Embedded engineer with hands-on experience in real-time sensor data processing, thermal imaging pipelines, and hardware-software co-design. Holding an Erasmus Mundus Master's degree in Smart Systems Integrated Solutions as a fully funded scholar, with academic training across Finland, Norway, and Hungary. Skilled in embedded C/C++, PCB design, experimental validation.

EXPERIENCE

Integrated Detector Electronics AS (IDEAS) Oslo, Norway
Intern *06/2025 – 08/2025 & 01/2026 – 06/2026*

- **Project.** *Agrarsense - **Master's thesis**: Embedded Calibration Storage and Drift Correction for an Uncooled Microbolometer Thermal Imaging System*
  - Developed a real-time Qt/C++ desktop application for a 640×480 LWIR thermal camera (Lynred ATTO640D-02 microbolometer), handling 16-bit raw video over RTP/UDP Ethernet using a patched FFmpeg backend; improved throughput from 9 to 45+ FPS via profiling-driven refactoring and SSE4-vectorized histogram computation.
  - Implemented GPU-accelerated DSP pipeline using custom GLSL shaders for Non-Uniformity Correction and Ironbow colorization, integrated into Qt's Scene Graph rendering framework.
  - Designed multi-threaded architecture with independent QThread consumers for real-time display, histogram/statistical analysis, and video recording with proper mutex synchronization.
  - Characterized microbolometer self-heating drift through a controlled measurement campaign; derived a per-row exponential correction model, validated it against experimental datasets from 2 camera units.
  - Worked on embedded C firmware for a Zynq-7000 platform (Trenz TE0720) running FreeRTOS, covering temperature-dependent bias control and on-device data handling.

Academic project – Erasmus Mundus SSIs Finland – Norway - Hungary
Master student *09/2024 – 12/2025*

- **Project.** *VitaCheck – Health Monitoring Device **(Startup Concept) – 1.5 years project.***
- Worked with 5 teammates to develop a device with 2 main functions: Hand grip strength measurement and finger tapping that can provide data for people to track their health parameters trending, focus on the elderly people, using Raspberry Pi 4 and 7 inch touch screen. Sensor for hand grip strength is MB Miniature Beam Load Cell combine with amplifier HX711. Then after, proposed a strategy to bring the device to the market also considering legality and competitiveness.
- Main responsibility was about the prototype design and fabrication. Software for 3D models is SolidWorks. Fabrication by 3D printer. Besides, researched and compared the characteristics and parameters of similar products on the markets.
- Worked on a smart blood glucose sensor system integrating electrochemical sensing, analog front-end, and digital signal processing using VHDL (Model Sim, Quartus) and ANSYS Twin Builder for system validation.
- Output: The project completed with a device works well with reasonable strategy for the market that secured successfully a hypothetical investment of $600,000 for 10% equity from 4 out of 5 evaluators in a mock crowdfunding evaluation.

FPT Japan Holdings Hanoi, Vietnam
Embedded engineer (Autosar) - On job training *02/2024 – 08/2024*

- Strengthened C++ programming skills through hands-on development of a multimedia management application on Ubuntu.
- Built foundational knowledge in embedded systems, including processors, interrupts, and communication protocols.
- Gained experience working in a structured engineering environment with emphasis on code organization, teamwork, and technical communication.

3CsLab Hanoi, Vietnam
Research student *12/2019 – 01/2024*

- Played a role as a key member to work and support with several projects, seminars.
- Managed to develop several projects related to robotics, IoT and mechatronics. Notable projects:
- **Project 1.** *IoT and Machine learning-based motor monitoring device:* *02/2023 – 08/2023*
  - Led a team of 5 members to develop a device mounted on motor/engine cases for monitoring motors' parameters and detecting the abnormal signal by using classification deep learning model, then send an alert to a mobile application.

- Worked on schematic and layout PCB using Altium Designer. Designed 3D model by SolidWorks. Then fabricated by hand-soldering components to the PCB and using 3D printer. Tested successfully 4 versions to give the most efficient and stable.
- Output: The device performed with accuracy over 95%. The project won the Second prize in the Scientific 2023 Research Conference by HUST among 33 other mechanics/mechatronics projects.

- **Project 2.** *Smart gate system:* *01/2022 – 06/2022*
  - Led a team of 3 members to develop an IoT-based auto-gate system, which is opened by face detection or RFID cards and controlled with an admin webpage which manages and visualizes information of all RFID tags.
  - Developed local webserver by PHP, HTML, JavaScript & database by using XAMPP, MySQL.
  - Designed a UID-reader device to take the new ID card using ESP32, RFID reader by SolidWorks, programed for it using C.
  - Output: The designed system had well-performed fully mentioned functions and won the first prize in the Scientific 2022 Research Conference organized for the 39th time by HUST among 34 other mechanics/mechatronics projects.

EDUCATION

Smart System Integrated solutions (SSIs) - Erasmus Mundus Joint Master's Degree **Finland – Norway – Hungary**
Master student 09/2024 – 08/2026

- Erasmus Mundus Scholar – fully funded
- Studied in University of South-Eastern Norway, Budapest University of Technology and Economics (BME) and Aalto University.
- Focus on Smart Systems Integration with a strong emphasis on design practice, combining theoretical and practical training in system-level design, IC design, VHDL-based hardware development, with MEMS fabrication and multiphysics modelling.

Hanoi University of Science and Technology (HUST) Hanoi, Vietnam
Mechatronics bachelor (Honors program) *9/2019 – 10/2023*

- #351-400 QS WUR Ranking by Subject in Engineering - Electrical and Electronic (topuniversities.com/universities/HUST).
- Thesis topic: **IoT and Machine learning-based motor monitoring device –** Grade: 9.5/10 or A+.

HONORS AND AWARDS

- Awarded **Erasmus Mundus Joint Master's degree Scholarship** in **Smart System Integrated solutions** program (2024).
- Accomplished **2nd place** at the 40th HUST (Hanoi University of Science and Technology) Scientific Research Conference with **IoT and Machine learning-based motor monitoring device** (2023).
- Accomplished **1st place** at the 39th HUST Scientific Research Conference with **Smart gate system** (2022).
- Accomplished **2nd place** at HUST Invention Exhibition Competition for Youth with **Smart gate system** (2022).
- Was selected one of 10 teams for final round at **BK – ROV (Remotely operated underwater vehicle)** competition and accomplished a **'Smartest product'** prize (2021).
- Was awarded **HUST scholarship** for excellent academic performance (2021).

SKILLS

Programming: C, C++ (modern, multi-threaded), VHDL, Python (data analysis).
Embedded & FPGA Platforms: ESP32, STM32, Arduino, Xilinx Zynq-7000 SoC
Imaging & DSP: Qt/QML, GLSL shaders, OpenGL, FFmpeg, SIMD/SSE4, non-uniformity correction, sensor calibration
Expertise: Embedded systems, real-time software, hardware-near programming, system integration, DSP, FPGA-based systems, hardware-software co-design, PCB design
Interfaces: UART, SPI, I2C, Ethernet (UDP/RTP)
Tools & Software: Altium Designer, ModelSim, ANSYS Twin Builder, MATLAB, SolidWorks, Vivado, Vitis, Visual Studio Code
Languages: Vietnamese (native), English (proficient – IELTS 7.0), Japanese (JLPT N3)
Additional Skills: Analytical thinking, problem-solving, fast learning, self-learning, teamwork, AI-assisted development

RESEARCH QUALIFICATIONS

1. Truong, D. P., **Nguyen, H. V**., & Dang, T. B. (2025). High Accurate Smart Device for Real-Time Monitoring Electric Motor Conditions Based on IoT Technology and Artificial Intelligence. JST: Smart Systems and Devices, 35(2), 25-34. https://doi.org/10.51316/jst.182.ssad.2025.35.2.4
2. Truong Duc Phuc, Tran Van Thanh, Le Tien Cuong, **Nguyen Hoang Vu**, Dang Tran Bach. "Development of smart disinfection machine using IoT technology" (2023). DOI : 10.17973/MMSJ.2023_10_2023057.
3. Phuc, T.D., **Vu, N.H**., Duong, H.D.T., Bach, D.T., Son, B.C. (2024). Development of IoT-Based Device for Nonintrusive Measurement of Rotational Speed of Induction Motors. In: Hai, N.T., Huy, N.X., Amine, K., Lam, T.D. (eds) EAI International Conference on Renewable Energy and Sustainable Manufacturing. EAI ICRESM 2023. EAI/Springer Innovations in Communication and Computing. Springer, Cham. https://doi.org/10.1007/978-3-031-60154-5_38 .

INTEREST

Reading book – Drawing & Painting – Swimming – Photography - Making Robot – Coffee

REFERENCE

Dirk Meier – CTO, Integrated Detector Electronics AS – Contact details available on request